

Örnekler:

$$1) 3 - 5 \cdot [2 - 4 - 8] \cdot 4 - 3 \cdot (45) - 2 - 5 \cdot (3 - 4) = ?$$

$$3 - 5 \cdot (2 + 7 \cdot 4 + 15) - 2 - 5 \cdot (-1)$$

$$3 - 5 \cdot (45) - 2 + 5 = 6 - 225 = -219$$

$$2) (-3)^2 - \left[\frac{(1-30)}{-10} \cdot 3 + \frac{2 \cdot (1-5)}{-10} - \frac{8 \cdot (2-5)}{24} \right] \cdot 2 - 9 = ?$$

$$9 - (-10 - 10 + 24) \cdot 2 - 9 = -4 \cdot 2 = -8$$

$$3) x - \{1 - [2x - 3 \cdot (x+1)]\} + 4 \cdot (2 - 2x) = ?$$

$$x - [1 - (2x - 3x - 3)] + 8 - 8x$$

$$x - 1 + 2x - 3x - 3 + 8 - 8x = 4 - 8x$$

$$4) -3 \cdot [(x+2y) - 3 - 2 \cdot (x+y)] - [(3x+1) - (x-8)] = ?$$

$$-3 \cdot (x+2y - 3 - 2x - 2y) - (3x+1 - x+8)$$

$$-3 \cdot (-3-x) - 3x - 1 + x - 8 = 9 + 3x - 3x - 1 + x - 8 = x$$

Kesirlerde Toplama ve Çıkarma:

$$-\frac{5}{6} + \frac{4}{5} + \frac{2}{3} = \frac{25+24+20}{30} = \frac{69}{30} \quad -5\frac{1}{2} - \frac{3}{5} = \frac{11}{2} - \frac{3}{5} = \frac{55-6}{10} = \frac{49}{10}$$

Çarpma:

$$-\frac{5}{8} \cdot \frac{3}{4} = \frac{15}{32} \quad -\frac{2}{3} \cdot 5\frac{4}{5} = \frac{2}{3} \cdot \frac{29}{5} = \frac{58}{15} \quad -2 \cdot \frac{3}{5} = \frac{6}{5}$$

$$-2\frac{3}{4} \cdot (-3\frac{5}{6}) \cdot \frac{24}{11} = \frac{11}{4} \cdot (-\frac{23}{6}) \cdot \frac{24}{11} = -23$$



Bölme

$$-\frac{3}{4} : \frac{2}{3} = \frac{3}{4} \cdot \frac{3}{2} = \frac{9}{8} \quad -2\frac{1}{5} : 1\frac{1}{3} = \frac{11}{5} : \frac{4}{3} = \frac{11}{5} \cdot \frac{3}{4} = \frac{33}{20}$$

Örnekler:

1) $(2\frac{1}{3} + 1\frac{1}{4}) - (1\frac{1}{2} - 1\frac{1}{6}) = ?$

$$\underset{(1)}{\frac{2}{3}} + \underset{(3)}{\frac{5}{4}} - (\underset{(2)}{\frac{3}{2}} - \underset{(4)}{\frac{5}{6}}) = \frac{28+15}{12} - \frac{6-5}{4} = \frac{43}{12} - \frac{1}{4} = \frac{43}{12} - \frac{3}{12} = \frac{40}{12} = \frac{10}{3}$$

2) $[-\frac{3}{4} : \frac{2}{3}] \cdot [\frac{5}{3} \cdot (-\frac{4}{3})] = ?$

$$(-\frac{3}{4} \cdot \frac{3}{2}) \cdot (-\frac{20}{9}) = (-\frac{9}{8}) \cdot (-\frac{20}{9}) = \frac{20}{8} = \frac{5}{2}$$

3) $(1\frac{2}{3})^{-1} + (\frac{2}{7})^{-1} = (\frac{5}{3})^{-1} + (\frac{2}{7})^{-1} = \frac{3}{5} + \frac{7}{2} = \frac{6}{10} + \frac{35}{10} = \frac{41}{10}$

4) $\frac{(\frac{1}{5} : \frac{1}{3}) + (\frac{3}{4} - \frac{1}{8})}{\frac{7}{8} : \frac{5}{7}} = ? \quad (\frac{1}{5} \cdot 3) + (\frac{6-1}{8}) = \frac{\frac{3}{5} + \frac{5}{8}}{\frac{7}{8} \cdot \frac{7}{5}} = \frac{\frac{24}{40} + \frac{25}{40}}{\frac{49}{40}} = \frac{49}{40} : \frac{49}{40} = 1$

5) $\left(\frac{1 - 1\frac{1}{3}}{1 + 1\frac{1}{3}}\right)^2 \cdot \left(\frac{7}{2}\right)^2 = \left(\frac{1 - \frac{4}{3}}{1 + \frac{4}{3}}\right)^2 \cdot \frac{49}{4} = \left(\frac{-\frac{1}{3}}{\frac{7}{3}}\right)^2 \cdot \frac{49}{4} = \left(\frac{1}{9}\right) \cdot \frac{49}{4} = \frac{49}{4}$

$$\frac{1}{9} \cdot \frac{49}{4} = \frac{49}{36}$$

6) $1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{3}}} = 1 + \frac{1}{1 + \frac{3}{4}} = 1 + \frac{4}{7} = \frac{11}{7}$

Globally $(1 + \frac{1}{3}) \rightarrow \frac{4}{3}$

7) $1 + \frac{\frac{1}{2} + \frac{2}{3\frac{1}{4}}}{1 + \frac{1}{1 + \frac{1}{2}}} = ?$

Ondalık Kesirler

$$\frac{2}{10} = 0,2 \quad \frac{3}{100} = 0,03$$

Toplama ve Çıkarma

$$-12,37 + 2,084 = -10,286$$

Çarpma

$$-36,5 \cdot 0,24 = -8,76$$

$$\frac{3}{4} = \frac{33}{20}$$

$$1 + \frac{\frac{1}{2} + \frac{2}{3/4}}{1 + \frac{1}{\frac{1}{2} + \frac{1}{2}}} = 1 + \frac{\frac{19}{6}}{1 + \frac{2}{3}} = 1 + \frac{\frac{19}{6}}{\frac{5}{3}} = 1 + \frac{19}{6} \cdot \frac{3}{5} = 1 + \frac{19}{10} = \frac{29}{10}$$

Ondalık Kesirler Örnekler

$$\frac{2}{10} = 0,2 \quad \frac{3}{100} = 0,03 \quad 4 \frac{75}{100} = 4,75 \quad \frac{125}{10} = 12,5 \quad \frac{375}{1000} = 0,375$$

Toplama ve Çıkarma

$$-12,37 + 2,084 = ? \quad -8,04 - 3,325 = ?$$

$$= 14,454 \quad = 4,715$$

Çarpma

$$-36,5 \cdot 0,24 = ?$$

$$\begin{array}{r} 36,50 \\ \times 0,24 \\ \hline 14600 \\ + 7300 \\ \hline 8760 \end{array}$$

Bölme

$$\frac{0,02}{0,004} + \frac{0,03}{0,005} + \frac{0,04}{0,08} = ?$$

$$\frac{\frac{2}{100}}{\frac{4}{1000}} + \frac{\frac{3}{100}}{\frac{5}{1000}} + \frac{\frac{4}{100}}{\frac{8}{100}}$$

$$\frac{2}{100} \cdot \frac{1000}{4} + \frac{3}{100} \cdot \frac{1000}{5} + \frac{4}{100} \cdot \frac{100}{8}$$

$$= 5 + \frac{30}{5} + \frac{1}{2} = 5 + 6 + \frac{1}{2} = \frac{11}{2}$$

$$= \frac{23}{2}$$

